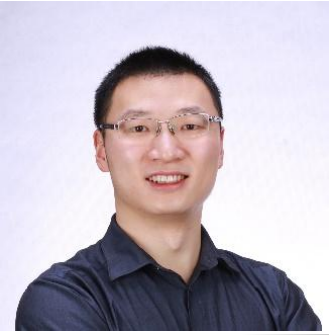
Zhe WANG

Associate Professor, Hong Kong University of Science and Technology
Associate Director, HKUST Smart City Institute
Email: cezhewang@ust.hk
Phone: (+852) 2358 8753
Website: https://walterzwang.github.io/
Address: Academic Building Room 3564, HKUST, Clear Water Bay, Hong Kong SAR



INTRO

I am an Associate Professor working with The Hong Kong University of Science and Technology. I am also serving as the Associate Director of HKUST Smart City Institute. My current research is about the application of artificial intelligence in smart building and low carbon city, majorly from two aspects. First, I am looking at how to leverage advanced control techniques (e.g. Reinforcement Learning, Model Predictive Control, etc.) to enhance building performance and building-grid interaction. Second, I am interested in how machine learning techniques could help to inform decision making on buildings and cities.

Prior to joining HKUST, I was a scientist in Lawrence Berkeley National Laboratory working on smart building control; a Postdoctoral Scholar in UC Berkeley working on data-driven personal comfort models and devices; and an Energy Consultant in World Bank working on Urban Scale Building Energy Efficiency and Renewable Energy in China.

I have been named as the World's Top 2% Scientists by Stanford University & Elsevier since 2021, for the fourth consecutive year. I was awarded the **National Science Fund for Excellent Young Scholars** (2023), **Hong Kong Collaborative Research Fund** (PC, 2023), **Second Class Award of State Science and Technology Prize** (2019); and was selected as a Schwarzman Scholar (2016, covered by the New York Times). I am now serving as a Section Editor for Energy and Buildings, Subject Editor for Building Simulation, and Assistant Editor for Advances in Applied Energy. I am a Fellow of the George H. W. Bush Foundation for U.S.-China Relations. I have obtained **14.8 million HKD** (**1.9 million USD**) external research funding as the PC or PI since I joined HKUST. I have won the first prize in the **2022 Global AI Challenge for Building E&M Facilities** hosted by Hong Kong Electrical & Mechanical Services Department, and the second prize in the **NeurIPS CityLearn Challenge 2023** (Leaderboard here). I was invited to give a speech on the Seventh-Round High-level Consultation on U.S.-China People to People Exchange, chaired by U.S. Secretary of States John Kerry and Chinese Vice-Premier Yandong Liu, as the ***only representative of Chinese young scholar*** (news coverage).

EMPLOYMENT

2021.11 till now **HKUST, Assistant Professor (21-25), Associate Professor (25 till now), Associate Director of HKUST Smart City Institute (25 till now), Management Committee Member of Hong Kong Space Robotics and Energy Center (26 till now)** **Hong Kong**

- PI, Key technologies for the efficient building-grid interaction and building load flexibility in hot and humid area (**2M RMB**, National Key R&D Program of China), 2024-2028
- PI, Advanced building control for energy efficiency and load flexibility (**2M RMB**, the National Science Fund for Excellent Young Scholars by NSFC, 52322813), 2024-2026
- PI, Semantic AI empowered fault detection and control optimization for HVAC system of large-scale commercial buildings (**900k HKD**, the Innovation and Technology Fund by ITC, PRP-061-23FX), 2025-2028
- PI, Cabin thermal management for electrical vehicles (**300k RMB**, the National Science Fund for Young Scholars by NSFC, 52306028), 2024-2026
- PI，Towards a human-centric smart cabin: System design integrating thermal comfort and air quality (**500k HKD**，HKUST – HKUST(GZ) 20 for 20 Cross-campus Collaborative Research Scheme，C006)，2024-2026

- PI, Enhancing Operational Efficiency and Reliability of Data Center (2**00k RMB**, Tencent Young Faculty Open Research Fund), 2024-2025
- PI, Smart algorithms and software for construction safety monitoring (**1M RMB**，Shenzhen Ridge Inc.), 2024
- PI, Smart Platform for Demand Response of Commercial Buildings - Period II (**600k RMB**, Midea Group), 2024
- PC, Toward 2060 Carbon Neutrality: Life-cycle Planning and Design of Photovoltaic Integrated Green Roof (PVIGR) Systems for Hong Kong and the Greater Bay Area (**3.46M HKD**, Hong Kong Research Grant Committee, Collaborative Research Fund, C6003-22Y), 2023-2026
- PI, Model Predictive Control in Hong Kong's Residential Buildings for Energy Efficiency and Load Flexibility (**804k HKD**, Hong Kong Research Grant Committee, Early Career Scheme, 26209323), 2023-2026
- PI, Fault detection and control optimization for air-conditioning system of commercial buildings (**800k RMB**, Shenzhen Sustainable Development Research Grant, KCXST20221021111403009), 2023-2026
- PI，An AI-assisted solution for low-cost high-resolution urban scale environmental simulation (**360k HKD**, Fei Chi En Education and Research Fund), 2023-2025
- PI, Modeling and optimization for building energy system, (**300k RMB**, Shui On Sustainable Research Fund), 2023-2025
- PI, Autonomous Cruise UVC Disinfection and Microclimate Air-conditioning Robot (**720k RMB**, SHCIRI-FSNH-2203), 2022-2024
- PI, Smart Platform for Demand Response of Commercial Buildings (**600k RMB**, Midea Group), 2022-2023
- PI, Model Predictive Control for Energy Efficient Data Center (**300k RMB**, Tencent Young Faculty Open Research Fund), 2022-2023

2018.07 – 2021.10 **Lawrence Berkeley National Lab, Project Scientist** **Berkeley**

- Co-PI, AlphaBuilding – Reinforcement Learning for Building Control (0.2M USD, LBNL LDRD: BU21-036)
  - Develop virtual testbeds to train and benchmark RL controllers for buildings
  - Design and implement multi-agent RL structure to enhance the scalability of RL controllers
- Leading researcher, Sensor Data Integration - Integrating Sensor Data with Physics-Based Models (1.5M USD, US Department of Energy: EE-5B37579)
  - Infer building thermal dynamics using connected smart thermostat data
  - Estimate demand response potential of US residential building stock
- Key researcher, Hierarchical Occupancy Responsive Model Predictive Control at Room, Building and Campus Levels (3.0M USD, US Department of Energy: EE-5B24502)
  - Develop, deploy, and test Model Predictive Control for buildings
  - Develop toolkits to simplify MPC controller development process
- Key researcher, End-Use Load Profiles for the U.S. Building Stock (1.2M USD, US Department of Energy: EE-5B35033)
  - Develop occupant behavior modelling toolkits to generate realistic building load shape
  - Develop and validate data-driven approach to generate building load using GAN

2017.10-2018.06 **UC Berkeley, Postdoc Researcher** **Berkeley**

- Key researcher, Data-driven personal comfort models and wearable comfort devices (0.3M USD, US National Science Foundation)
  - Develop data-driven models to predict individual comfort demands
  - Develop and test energy-efficient personal comfort wearables to address individualized demands

2016.12-2018.06 **World Bank, Energy Consultant** **Beijing/Berkeley**

- Joint research with Ministry of Housing, Urban and Rural Development: Urban Scale Building Energy Efficiency and Renewable Energy Project
  - Assist the project manager by overseeing the technical progress
  - Provide technical supports to the sub-contractors

EDUCATION

2011.08-2017.07 **Tsinghua University** ***Ph.D. in Building Science***

- Visiting scholar in ***Harvard University***, 2015
- Recipient of the *National Scholarship* (2012), *Boeing Scholarship* (2015), *Outstanding Graduate of Beijing* (2017)

2016.08-2017.07 **Schwarzman College, Tsinghua University** ***Master in Public Policy***

- Invited to give a speech on the Seventh-Round High-level Consultation on U.S.-China People to People Exchange (Chaired by Chinese Vice Premier Yandong Liu and U.S. Secretary of State John Kerry), as the *ONLY* representative of Chinese young scholar
- Selected as a Student Council Member (12 out of 110)

2013.10-2014.09 **University of Cambridge** ***M.Phil. in Energy Technology***

- Recipient of the *Wing Yip Scholarship, Cambridge Overseas Trust* (*2013*)

2008.09-2011.07 **Tsinghua University** ***Bachelor in Economics***

- Cumulative GPA 92.8/100, **ranking 1st** /126

2007.09-2011.07 **Tsinghua University** ***Bachelor in Civil Engineering***

- Cumulative GPA 91.3/100, **ranking 1st** /29
- Recipient of the *National Scholarship* (2009, 2010), *Boeing Scholarship* (2011)
- Recipient of the *Outstanding Undergraduate of Beijing* (2011)

OPEN-SOURCE TOOL

MPCPy	Python-based open-source platform for model predictive control in buildings https://github.com/lbl-srg/MPCPy
Modelica Buildings Library	Dynamic simulation models for building energy and control systems https://github.com/lbl-srg/modelica-buildings

ACADEMIC SERVICE

Journal Editor	*Energy and Buildings,* **Section Editor** *in Intelligent Building and Smart Control* *Building Simulation*, **Subject Editor** in *Building Control* *Advances in Applied Energy,* **Assistant Editor**
Administration	Associate Director of HKUST Smart City Institute Member of the University Committee on Student Affairs
Grant Proposal Reviewer	*Natural Sciences and Engineering Research Council Grant*, Canada, 2025 *Frontier Competitive Research Program*, Singapore, 2024 *MTR Research Funding Scheme, HK, 2023* *The Dunhill Medical Trust, UK, 2020*
Conference Organizing	*Asian University Alliance Academic Conference on smart building*, **Conference Chair**, 2024 *Jiangsu-Hong Kong-Macau University Alliance Seminar on smart city and green building*, **Conference Chair**, 2023 *International Workshop on Reinforcement Learning for Energy Management in Buildings & Cities (RLEM)*, https://rlem-workshop.net/, **Member of Technical Program Committee**, 2020, 2021, 2023

	The 13th *International Symposium on Heating, Ventilation and Air-conditioning (ISHVAC)*, http://ishvac2023.org/, **Member of Scientific Committee**, 2023
Journal Reviewer	*Nature Energy, Applied Energy, Energy, Building and Environment, Energy and Building, Journal of Building Engineering, Building Simulation, Environmental Science and Pollution Research, Science and Technology for the Built Environment, SoftwareX, Journal of Building Performance Simulation, Applied Thermal Engineering, Advanced Engineering Informatics, Frontiers in Built Environment, International Journal of Refrigeration, International Journal of Biometeorology, Journal of Asian Architecture and Building Engineering, Journal of the Taiwan Institute of Chemical Engineer, Sustainable Cities and Society, Engineering*

AWARD

1. Key Technology for the Environmental and Energy-Efficient Design of Green Building, Second Class Award of 2019 Chinese State Science and Technology Prize, Award Number: 2019-J-22101-2-01-R09
2. First Prize, 2022 Global AI Challenge for Building E&M Facilities, hosted by Hong Kong Electrical and Mechanical Services Department (EMSD)
3. Second Prize, 2023 NeurIPS CityLearn Challenge, control track, hosted by AIcrowd, UT Austin, and NeurIPS
4. Key Technology for the Environmental and Energy-Efficient Design of Green Building, First Class Award of 2018 Beijing Science and Technology Prize
5. Environment Monitoring and Energy Conservation Technology of Large Commercial Buildings, First Class Award of 2020 Huaxia Construction Prize

PATENT

1. **Wang, Z.**, A Smart Thermostat for Demand Response in Residential Buildings. Chinese Patent Application Number: 202210667572.X
2. **Wang, Z**., Zhang, S., Chen, L., Wu, Z., Li, L., Wang, C., Liu, W., Automatic modelling method for urban scale building energy simulation, 202410909228.6
3. **Wang, Z**., Li, S., Li, S.H., Optimization method, system and product for chiller sequencing, 202410909230.3
4. **Wang, Z**., Zhou, Q., Zheng, Y., Path planning for mobile sensing of indoor air quality, 202410909229.0
5. Lin, BR., **Wang, Z.**, Li, ZW., Pei, ZF., Liu, K., Energy Efficient Air Conditioning System for Residential Buildings. Chinese Patent Number: ZL 201310161191.5
6. Lin, BR., Zhao, HT., and **Wang, Z.**, A Waste Heat Recovery System Designed for Dishwasher. Chinese Patent Number: ZL 201320328067.9

PUBLICATION

Google Scholar Citation: 10234; h-index: 52; i10-index: 102; ESI Highly Cited papers: 6

(#: contribute equally; *: corresponding author)

First Author

1. **Wang, Z.*** and He, Y., 2023. AlphaHydrogen: A virtual platform for simulating and evaluating station-based regional hydrogen-electricity networks with distributed renewables, buildings, and fuel-cell vehicles. Energy Conversion and Management, 280, p.116802.
2. **Wang, Z.***, 2022. How frequent should we measure the indoor thermal environment. Building and Environment, 222, p.109464.
3. **Wang, Z.**, Chen, B., Li, H. and Hong, T., 2021. AlphaBuilding ResCommunity: A multi-agent virtual testbed for community-level load coordination. Advances in Applied Energy, 4, p.100061.
4. **Wang, Z.**, Hong, T. and Li, H., 2021. Informing the planning of rotating power outages in heat waves through data analytics of connected smart thermostats for residential buildings. Environmental Research Letters, 16(7), p.074003.
5. **Wang, Z.**, Hong, T., Li, H. and Piette, M.A., 2021. Predicting city-scale daily electricity consumption using data-driven models. Advances in Applied Energy, 2, p.100025.
6. **Wang, Z.** and Hong, T., 2020. Reinforcement Learning for Building Controls: The opportunities and challenges. Applied Energy, 269, p.115036. (highly cited paper)

7. **Wang, Z.**, Hong, T. and Piette, M.A., 2020. Building thermal load prediction through shallow machine learning and deep learning. Applied Energy, 263, p.114683. (highly cited paper)
8. **Wang, Z.** and Hong, T., 2020. Learning occupants' indoor comfort temperature through a Bayesian inference approach for office buildings in United States. Renewable and Sustainable Energy Reviews, 119, p.109593.
9. **Wang, Z.** and Hong, T., 2020. Generating realistic building electrical load profiles through the Generative Adversarial Network (GAN). Energy and Buildings, p.110299.
10. **Wang, Z.**, Hong, T., Piette, M.A. and Pritoni, M. 2019, Inferring occupant counts from Wi-Fi data in buildings through machine learning, Building and Environment, 158, pp. 281-294.
11. **Wang, Z.**, Parkinson, T., Li, P., Lin, B. and Hong, T., 2019. The Squeaky wheel: Machine learning for anomaly detection in subjective thermal comfort votes. Building and Environment, 151, pp.219-227.
12. **Wang, Z.**, Zhang, H., He, Y., Luo, M., Li, Z., Hong, T. and Lin, B., 2020. Revisiting individual and group differences in thermal comfort based on ASHRAE database. Energy and Buildings, 219, p.110017.
13. **Wang, Z.**, Wang, J., He, Y., Liu, Y., Lin, B. and Hong, T., 2020. Dimension analysis of subjective thermal comfort metrics based on ASHRAE Global Thermal Comfort Database using machine learning. Journal of Building Engineering, 29, p.101120.
14. **Wang, Z.**, Hong, T. and Piette, M.A., 2019. Predicting plug loads with occupant count data through a deep learning approach. Energy, 181, pp.29-42.
15. **Wang, Z.**, Warren, K., Luo, M., He, X., Zhang, H., Arens, E., Chen, W., He, Y., Hu, Y., Jin, L. and Liu, S., 2019. Evaluating the comfort of thermally dynamic wearable devices. Building and Environment, p.106443.
16. **Wang, Z.**, Hong, T. and Piette, M.A., 2019. Data fusion in predicting internal heat gains for office buildings through a deep learning approach. Applied Energy, 240, pp.386-398.
17. **Wang, Z.**, Hong, T. and Jia, R., 2018. Buildings. Occupants: a Modelica package for modelling occupant behaviour in buildings. Journal of Building Performance Simulation, pp.1-12.
18. **Wang, Z.**, Luo, M., Geng, Y., Lin, B. and Zhu, Y., 2018. A model to compare convective and radiant heating systems for intermittent space heating. Applied Energy, 215, pp.211-226.
19. **Wang, Z.**, de Dear, R., Luo, M., Lin, B., He, Y., Ghahramani, A. and Zhu, Y., 2018. Individual difference in thermal comfort: A literature review. Building and Environment, 138, pp. 181-193 (highly cited paper)
20. **Wang, Z.**, Zhao, Z., Lin, B., Zhu, Y. and Ouyang, Q., 2015. Residential heating energy consumption modeling through a bottom-up approach for China's Hot Summer–Cold Winter climatic region. Energy and Buildings, 109, pp.65-74.
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22. **Wang, Z.**, de Dear, R., Lin, B., Zhu, Y. and Ouyang, Q., 2015. Rational selection of heating temperature set points for China's hot summer–Cold winter climatic region. Building and Environment, 93, pp.63-70.
23. **Wang, Z.**, Lin, B. and Zhu, Y., 2015. Modeling and measurement study on an intermittent heating system of a residence in Cambridgeshire. Building and Environment, 92, pp.380-386.

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24. Liu, S.*[,#], **Wang, Z.***[,#], Schiavon, S., He, Y., Luo, M., Zhang, H. and Arens, E., 2020. Predicted percentage dissatisfied with vertical temperature gradient. Energy and Buildings, p.110085.
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27. Liu, Y.[#], **Wang, Z.**[#], Lin, B., Hong, J. and Zhu, Y., 2018. Occupant satisfaction in Three-Star-certified office buildings based on comparative study using LEED and BREEAM. Building and Environment, 132, pp.1-10.
28. Touzani, S.[#], Prakash, A.K.[#], **Wang, Z.**[#], Agarwal, S., Pritoni, M., Kiran, M., Brown, R. and Granderson, J., 2021. Controlling distributed energy resources via deep reinforcement learning for load flexibility and energy efficiency. Applied Energy, 304, p.117733.

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32. Li, L., Pu, Y., Ju, Y., Hu, Z. and **Wang, Z.***, 2026. Coordinating Multiple UPS Batteries in Datacenter for Load Flexibility and Cost Reduction. Energy and Buildings, p.117282.
33. Li, M., Hu, Z., Mohebi, P., Li, S. and **Wang, Z.***, 2026. Enhancing LLM-based building data query with chain-of-thought, retrieval-augmented generation, and fine-tuning. Automation in Construction, 182, p.106738.
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36. Huang, Q. and **Wang, Z.***, 2026. Towards energy-positive and eco-profitable future: A perspective on building-to-vehicle-to-building ecosystems. Sustainable Energy Technologies and Assessments, 85, p.104760.
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38. Wang, Z., Sheng, H., Gu, F., Zhou, Y., Zhao, L., **Wang, Z.*** and He, D., 2025. Investigating effects of temperature and CO2 on driver drowsiness in the context of conditional automated driving. Ergonomics, pp.1-18.
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51. Wu, S., Zheng, W., **Wang, Z.***, Chen, G., Yang, P., Yue, S., Li, D. and Wu, Y., 2025. AlphaDataCenterCooling: A virtual testbed for evaluating operational strategies in data center cooling plants. Applied Energy, 380, p.125100.

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56. Wang, D., Zheng, W., **Wang, Z.***, Huang, Y., Li, S., Li, D., Li, B. and Yan, R., 2025. Coordinating variable refrigerant flow system for effective demand response in commercial buildings. Energy and Buildings, p.115066.
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65. Wu, Z., **Wang, Z.***, Cheng, J.C. and Kwok, H.H., 2024. A knowledge-informed optimization framework for performance-based generative design of sustainable buildings. Applied Energy, 367, p.123318.
66. Wang, D., Zheng, W., **Wang, Z.***, Wu, Z., Shen, B. and Tian, S., 2024. Quantifying the potential of load flexibility for building HVAC system using model predictive control strategy. Energy and Buildings, p.114819.
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69. Zhao, L., Zhou, Q. and **Wang, Z.***, 2024. A systematic review on modelling the thermal environment of vehicle cabins. Applied Thermal Engineering, p.124142.
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System for Control Purpose. Energy and Buildings, p.113163.

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77. Ju, Y., **Wang, Z.***, Ju, X., Cao, B., Chen, C. and Lin, B., 2023. Understanding occupancy patterns of university libraries in the post-pandemic era. Energy and Buildings, 291, p.113138.
78. Wu, Z., Cheng, J.C., **Wang, Z.*** and Kwok, H.H., 2023. An ontology-based framework for automatic building energy modeling with thermal zoning. Energy and Buildings, p.113267.
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